Module 1 — Introduction & Environment

Week 1 · ~8 hours

1.1 What is sPHENIX, and what are we simulating?

sPHENIX is a mid-rapidity collider experiment at the Relativistic Heavy Ion Collider (RHIC) at Brookhaven National Laboratory, built around the former BaBar 1.5 T superconducting solenoid. It combines a silicon tracking system, an electromagnetic calorimeter (EMCal), and inner / outer hadronic calorimeters (IHCal, OHCal) to measure jets, upsilon states, and heavy-flavor hadrons in Au+Au, p+Au, and p+p collisions at $\sqrt{s_{NN}}$ = 200 GeV.

Physics goals depend on precise simulation: the response of the calorimeters to jets, the tracking efficiency for low-pT particles, and the embedding of signal events into realistic Au+Au backgrounds. This is where Pythia, HIJING, and Geant4 come in.

1.2 The Monte Carlo chain, conceptually

A full simulation proceeds in three stages:

- **Event generation.** A physics-motivated Monte Carlo produces a list of particles with 4-momenta and vertex positions. For p+p we use **Pythia 8**. For heavy-ion (A+A), we use **HIJING**, which is itself a heavy-ion wrapper that calls hard-scattering physics similar to Pythia.
- **Detector simulation.** Each stable particle is transported through a geometric model of the detector (magnetic field, materials, sensitive volumes) by **Geant4**. Energy deposits ("hits") are recorded in every sensitive volume traversed.
- **Digitization & reconstruction.** Hits are converted to electronic signals, clustered, and passed to reconstruction algorithms (tracking, calorimeter clustering). In sPHENIX this is handled by the Fun4All framework.

> **Scope:** For this course we go through stages 1 and 2 end-to-end. Digitization and reconstruction are briefly introduced in Module 8 but are not required for the capstone example.

1.3 Units and conventions

The HEP community uses natural units with ■ = c = 1 and energies in GeV. Geant4 internally uses CLHEP units (MeV, mm, ns) but accepts explicit units in all user code (e.g., 1.*CLHEP::GeV). Pythia uses GeV/mm/mm/c. HIJING uses GeV/fm. Be careful at boundaries; a consistent convention in your code will save hours of debugging.

The sPHENIX coordinate system: z is the beam direction, y is up, ϕ is the azimuthal angle in the x-y plane. Pseudorapidity is $\eta = -\ln \tan(\theta/2)$.

1.4 Reading for the week

- sPHENIX TDR chapter 1 (overview) — skim for vocabulary.
- Sjöstrand, arXiv:2203.11601 — sections 1 and 2 (overview of Pythia's event structure).

- Geant4 Book For Application Developers, chapter 1 (“Introduction”).

1.5 Hands-on: verify your environment

Complete the Software Setup Guide for your platform, then verify:

```
$ root -l -b -q -e 'cout<<"ROOT "<<gROOT->GetVersion()<<endl;'
$ pythia8-config --version
$ geant4-config --version
$ which hijing # or ls $HOME/sw/bin/hijing
```

Then run the Pythia quick-test:

```
$ cat > pytest.cc <<'EOF'
#include "Pythia8/Pythia.h"
int main() {
Pythia8::Pythia p;
p.readString("Beams:eCM = 200.");
p.readString("SoftQCD:all = on");
p.init();
int n = 0;
for (int i=0; i<10; ++i) if (p.next()) ++n;
std::cout << "Generated " << n << " events\n";
return 0;
}
EOF
$ g++ pytest.cc $(pythia8-config --cxxflags --ldflags) -o pytest
$ ./pytest
```

1.6 Self-assessment

- [ ] I can describe the three-stage MC chain in my own words.
- [ ] root, pythia8-config, geant4-config, and hijing all respond to --version or equivalent.
- [ ] I compiled and ran pytest.cc and saw “Generated 10 events”.
- [ ] I know what GeV, MeV, and fm are, and which tool expects which.